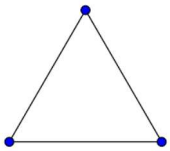
Problem 01: Circuit Design

Time limit: 18 seconds

Imagine you are an electronic engineer working on optimizing digital circuit layouts for integrated chips. Each circuit is represented as a graph where nodes correspond to components (like logic gates) and edges represent the connections between them.

Your task is to identify how often a specific subcircuit also called query subcircuit (a small group of components with a defined topology) appears in the overall circuit. Subcircuits can overlap, sharing components or connections, but each occurrence must have at least one unique component.

Given an undirected graph representing the circuit (components as nodes and connections as edges), identify and count all occurrences of the following pattern within the larger graph.



Input:

The input consists of the following:

- First line contains a pair ***N***, ***M***. The number of components in the circuit, $N\ (1 \le N \le 2000)$. The number of edges in the circuit, $M\ (0 \le M \le (N(N-1)/2))$.
- M lines, each containing two component IDs ***u*** and ***v*** $(0 \le u, v \le N-1)$, representing an undirected edge from component ***u*** to component ***v*** in the circuit.

Output:

Output a single integer, the count of occurrences of the query sub circuit within the provided overall circuit.

Sample Input	**Sample Output**
6 9 0 1 1 2 1 3 2 3 3 4 4 0 2 5 5 0 5 1	3